NEIL BURT MARTÍN

Mechatronics Engineer | Robotics & Control Systems

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PERSONAL STATEMENT

Mechatronics graduate with hands-on experience in robotics, control systems, and machine learning. Designed PID controllers and flight simulations for autonomous drones, developed reinforcement learning pipelines, and built a domain-specific LLM for embedded systems as a graded thesis (10/10). Experienced in international academic environments, including an exchange semester in Shanghai. Proficient in Python, MATLAB, C/C++, and Simulink. Currently leading the control and programming area of SDU's Micromouse Robotics Club.

EDUCATION

Bachelor of Engineering in Mechatronics (Diploma Engineer) 2022 – 2026

SDU Syddansk Universitet, Sønderborg, Denmark

Exchange Semester — SSPU Shanghai Second Polytechnic University

September 2024 – January 2025 | Coursework: Statistics, Probability, Project Management | Began studying Mandarin (HSK2)

THESIS

Fine-tuning Large Language Models for Specialized Assistance in Embedded Systems Engineering

Grade: 10/10

SDU Syddansk Universitet | *Supervisor: Benaoumeur Senouci*

- Fine-tuned Mistral 7B using Low-Rank Adaptation (LoRA) with a custom 1,651-sample domain-specific dataset
- Iterated through 9 training cycles to balance overfitting and underfitting; deployed via Ollama in GGUF format
- Evaluated against base model through structured user experiments; fine-tuned model showed consistent improvements in task completion rate, response clarity, and user confidence
- Model published on HuggingFace: huggingface.co/Neilburt/embbeded_assistant

UNIVERSITY PROJECTS

Drone Control Project — 4th Semester SDU, Sønderborg

- Designed, programmed, and simulated a drone capable of stable autonomous hovering
- Modelled the drone's plant system in Simulink and developed a PID controller for stable flight
- Applied control theory and simulation tools to ensure safe and reliable testing

Micromouse Robotics Club — Control & Programming Lead

SDU, Sønderborg | *Ongoing*

- Leading a team of 5 members in the design of control algorithms and software for autonomous maze navigation
- Coordinating development tasks and guiding implementation toward competition-ready performance

AWARDS & COMPETITIONS

HackKosice Hackathon — 1st Place April 2026

AT&T Challenge Winner • Best Use of ElevenLabs Award

- Developed Nabu, a voice-based technology assistant designed for elderly users, in a team of three
- Responsible for ElevenLabs voice API integration, frontend development, and part of the backend authentication
- Stack: Gemini API, ElevenLabs, Digital Ocean

WORK EXPERIENCE

Junior Engineer — Ruybesa Global Technologies March 2025 – July 2025

- Supported technical proposals and delivery of tailored engineering solutions in an international environment

SKILLS

Programming: Python, MATLAB, C/C++

Software & Tools: Simulink, NX Siemens, ANSYS, Git, Ollama, HuggingFace
Engineering Areas: Control systems, Robotics, Reinforcement learning, Embedded systems, Simulation
Languages: Spanish (Native), English (C1), Chinese (HSK2), Danish (Basic)